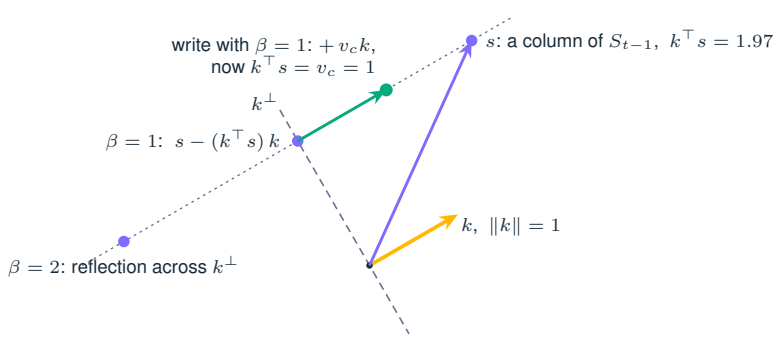


The transition $A_t = I - \beta_t k_t k_t^\top$

$A_t k_t = (1 - \beta_t)\, k_t, \quad A_t x = x$ for $x \perp k_t$

eigenvalues $1 - \beta_t$ (once) and 1,
so $\|A_t\|_2 = 1$ for $\beta_t \in [0, 2]$ and $r \geq 2$

$\beta_t = 1$: the orthogonal projection onto $k_t^\perp$
$\beta_t = 2$: the Householder reflection

The write, column by column

$s \leftarrow A_t s + \beta_t v_{t,c}\, k_t$ moves s along k_t only

with $\beta_t = 1$, it sets $k_t^\top s = v_{t,c}$: one step of the Kaczmarz method for the system $Ks = V_{\cdot c}$